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Marketing analytics capability, artificial intelligence adoption, and firms' competitive advantage: Evidence from the manufacturing industry

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Marketing analytics capability, artificial intelligence adoption, and firms' competitive advantage: Evidence from the manufacturing industry

Abstract

Data-driven analytics and artificial intelligence (AI) have become the most crucial aspects of today's industrial marketing management. Although many firms have embraced analytics and AI strategies, corresponding academic advances have been slow. This research investigates how industrial goods manufacturers sustain their competitive advantage in export markets, convincing buyers in a competitive data-rich business environment. The evidence has been taken from the RMG (readymade garment) industry, one of the largest manufacturing industries significantly attached to the export markets. Utilizing multi-phase research design, the study reveals that firms marketing analytics capability play a vital role in sensing, seizing, and reconfiguring the market, consequently leading to a sustained competitive advantage. The performance of sensing, seizing, and reconfiguring becomes higher for a firm when they adopt AI on the strength of the marketing analytics platform. These findings exhibit the latest avenue of exploration within marketing analytics and AI's academic research paradigm. Further, in practice, managers will be aware of the facts that create resilience in this specific industry context.

Keywords: Marketing analytics capability, AI adoption, Market sensing, Market seizing, Market reconfiguration, Manufacturing industry, B2B export market

1. Introduction

"Prior to the COVID-19 pandemic's stranglehold on the world, leaders increasingly embraced advanced analytics and artificial intelligence (AI) for a good reason. These capabilities are expected to offer annual economic value between $9.5 trillion and $15.4 trillion. Organizations are standing up analytics capabilities in a matter of weeks to inform business responses to COVID-19 challenges and prepare for the future"… McKinsey&Company (2020).

A business without data is now unimaginable. Data acts as 'the oil' for the digital economy since businesses generate meaningful insights from enormous data streams utilizing the power of marketing analytics (Wedel & Kannan, 2016). Notably, scholars from the marketing domain frequently highlight the appropriate use of analytics for tackling the disruption in marketing science and practice (Moorman, 2016; Urban, Timoshenko, Dhillon, & Hauser, 2020). Overall, marketers can use marketing analytics to decide how to use their marketing resources efficiently for business operations, find and keep customers or buyers who are likely to be profitable, and balance demand and supply coordination (Rahman, Hossain, & Fattah, 2021a). Marketing analytics is vital in shaping businesses to reach a new height. Researchers have also significantly contributed to providing directions in the academic literature on how marketing analytics impacts firms to drive their businesses and achieve long-term advantages in a B2C context (Addo, Akpatsa, Nukpe, Ohemeng, & Kulbo, 2022; Song, Zheng, Zhang, & Yu, 2018). However, research on marketing analytics capability is limited in the pure B2B manufacturing firm context.

Evidence suggests that manufacturing industries are fairly involved in exporting goods (Moktadir, Rahman, Rahman, Ali, & Paul, 2018). Readymade garment (RMG) has been categorized as one such manufacturing industry (Swazan & Das, 2022). The global RMG market size was valued at $983.7 billion in 2019 (Kumar & Deshmukh, 2020). Notwithstanding the ubiquitous importance of data analytics in RMG manufacturing firms, there is a knowledge gap in how analytics capability, specifically marketing analytics capability, can drive competitive advantage for such industries when producing products for buyers of export markets (Akter, Hossain, Lu, & Shams, 2021a; Ali et al., 2021; Chaudhary, Kumar, & Johri, 2020; Swazan & Das, 2022). These industrial challenges are essential to address as RMG manufacturers drive the global apparel industry that generates 2% of global GDP.

Further, evidence suggests both AI and analytics are transforming the US$3 trillion apparel industry at every level, from designing, producing, and logistics to marketing and sales (Forbes, 2019; Ravi, 2021). Data is the primary source for utilizing AI and analytics. Practitioners and scholars recognize such data-driven advancement as the base of innovation (Akter et al., 2020b; Zhang & Xiao, 2020), a source of market success (Cao et al., 2019; Davenport et al., 2020; Hossain et al., 2021), an engine of the organization (Davenport & Ronanki, 2018; Rahman et al., 2021a), and the catalyst to solve the business woes (Delen & Ram, 2018; Yalcin et al., 2022).

In the academic literature, a handful of studies on the apparel industry acknowledge AI and analytics separately from a B2C aspect (Giri, Thomassey, & Zeng, 2019; Silva, Hassani, & Madsen, 2019; Tupikovskaja-Omovie & Tyler, 2021). Overall, there is a lack of empirical evidence to guide manufacturing firms purely involved in B2B. In the B2C context, there have been scholarly discussions involving Big Data analytics capability (Akter, Wamba, Gunasekaran, Dubey, & Childe, 2016; Mikalef, Krogstie, Pappas, & Pavlou, 2019), Business analytics capability (Cosic, Shanks, & Maynard, 2015), Marketing analytics capability (Rahman et al., 2021a), and Customer analytics capability (Hossain et al., 2021). However, there is a knowledge gap in whether this same technique is applicable for manufacturing firms as most of their nature is purely B2B in the current export market. For example, in traditional RMG manufacturing business settings, producers used to get buyers' orders directly. But due to the recent progress of AI and analytics, most producers maintain ultimate customers' profiles (McKinsey&Company, 2020; Uddin, 2018). Thus, the manufacturing firms have many opportunities to convince the buyers by taking the actual scenario from the respective export markets. Despite such an option, the academic literature lacks conceptualization and empirical evidence from the context of B2B export-oriented manufacturing firms that regularly practice analytics and AI.

To fill the research gap, this study builds on the resource-based view (RBV) and Dynamic capability (DC) theory. Utilizing VRIO (valuable, rare, inimitable, and organized) resources, a firm generally offers its buyers uniquely and secures a competitive advantage (Akter, Gunasekaran, Wamba, Babu, & Hani, 2020a). However, critiques suggest versatile resources are crucial as VRIO limits the opportunities for proper deployment of resources (Hossain et al., 2021). Thus, dynamic capability theory is vital to managing versatile resources and tackling the gap of RBV (Gupta, Drave, Dwivedi, Baabdullah, & Ismagilova, 2020; Wang & Kim, 2017). Generally, RBV focuses on the firm's operational capabilities and existing tangible and intangible resources; dynamic capabilities focus on the most significant modification within its existing resource base to make firms more agile and ensure a competitive advantage (Mikalef et al., 2019; Randhawa, Wilden, & Gudergan, 2021). Theoretically, this study assumes marketing analytics capability as a firm's dynamic capability that can be strengthened further by adopting AI. Overall, dynamic capability act as a mechanism of potentially empowering some of the drawbacks of the RBV, where dynamic capability helps sense, seize, and reconfigure the firm's process to ensure benefits and competitive advantage (Endres, Helm, & Dowling, 2020; Lütjen, Schultz, Tietze, & Urmetzer, 2019). By considering the arguments, this research addresses the following research question.

RQ: How does the marketing analytics capability of an export-oriented manufacturing firm assure sustained competitive advantage?

This research comprehensively contributes to both theory and practice. Based on the theoretical views of RBVs and dynamic capabilities, this study improves our understanding of the instruments of marketing analytics capability in the B2B firm context. Besides, the theoretical extension shows how marketing analytics capability helps sense, seize, and reconfigure the market to gain sustained competitive advantage, where AI plays a vital role. Therefore, the research explicitly contributes to the analytics and AI literature by deepening our understanding of the complicated relations between marketing analytics capabilities and AI. Earlier studies narrowly focused on analytics capability and its direct influence on performance (e.g., Akter et al., 2016; Mikalef, Framnes, Danielsen, Krogstie, & Olsen, 2017; Mikalef et al., 2019; Wamba et al., 2017) and AI capability and its direct effect on performance (e.g., Davenport et al., 2020; Farrokhi, Shirazi, Hajli, & Tajvidi, 2020; Fethi & Pasiouras, 2010). There is a knowledge gap regarding how analytics capability can be accelerated using AI to sense, seize and reconfigure the market. The study shows the impact of AI adoption on a firm's analytics capability platform, which brings the latest avenues about how the paradigm of Analytics and AI research can progress and move forward in the export-oriented manufacturing firm context. Overall, these theoretical contributions shed new light and enhance our knowledge of existing B2B marketing research. In practice, managers of striving export-oriented manufacturing firms will know the fundamental mechanisms of marketing analytics capability and AI adoption from the research findings. In a more granular way, managers will learn how to transform efforts into marketing analytics actions, sync AI with marketing analytics strategies, and create pathways for anticipating sustained competitive advantage. Further, managers will be able to connect sustainable development goals (SDGs), especially *SDG 9 - Industry, Innovation, and Infrastructure,* utilizing these research findings.

The following section two verifies the problem and underpins the theories in detail to progress this research. Section three discusses the literature review, develops hypotheses, and conceptualizes the research model. Section four presents model validation methods and analysis. Section five discusses the research findings, research contributions and addresses limitations and future research directions. Section six concludes the study by drawing an overview of the research.

2. Theoretical background, study urgency, and problem verification

2.1. Firm's resource-based and dynamic capability views

Competitive advantage gives a business to leverage it needs to differentiate itself from competitors by exceeding buyers' expectations with value. When a business achieves a sustainable competitive advantage, its products, services, and brand value become distinctive. The resource-based view (RBV) portrays how a firm can optimally use its resources, which are non-substitutable, unique and valuable for achieving sustained competitive advantage (Wernerfelt, 1984). The seminal work of Wernerfelt (1984) is considered the pioneer study that contributed to formulating the RBV concepts, and other researchers led the baton to shape RBV as a complete resource-based theory (Kozlenkova, Samaha, & Palmatier, 2014). RBV offers theoretically supported guidelines to determine the firm's underlying resources and capabilities essential to achieving competitive advantages (Barney, 1991). RBV highlights the uttermost significance of a firm's resources and capabilities to enhance and optimize a firm's performance (Capron & Hulland, 1999). The core concept of RBV is the differentiation that a firm builds over time with its resource heterogeneity, possession of resources, and use of available resources and environmental opportunities to deliver distinctive products or services (Gaudenzi, Mola, & Rossignoli, 2021; Pereira & Bamel, 2021; Safari & Saleh, 2020). Due to the intense competition and challenges in the business environment, firms face numerous uncertainties and complexities in achieving and maintaining competitive advantages (Elia, Giuffrida, Mariani, & Bresciani, 2021). Since firms seek to gain a competitive advantage in the long run, it will depend on their ability to create new resources and capabilities, which are also incomparable to the competitors.

Despite the explanation in RBV of how firm resources contributes toward firms' performance differentiation, critics asserted that the broad definition of firm resources ignores the critical differences between assets and capabilities (Finney, Lueg, & Campbell, 2008). Therefore, academic scholars elaborated on the distinction between a firm's resources and capabilities (Greenley, Hooley, & Rudd, 2005; Nath, Nachiappan, & Ramanathan, 2010). Firm resources are tangible (property, plant, equipment, etc.) and intangible (information, knowledge, technology, processes, etc.) components. Firms use these resources to increase efficiency and effectiveness to implement organizational strategies to conquer the market competition (Amit & Schoemaker, 1993; Barney, 1991). Capability refers to the ability of the firm to use available resources to build invisible assets over time that cannot be imitable easily (Amit & Schoemaker, 1993). According to Auh and Menguc (2009), resources need to be nurtured to cope with transformation and reinvention and other complementary resources to prolong and protect their effectiveness, putting competitors in a defensive and follower's mode to do business in the market. Thus, it portrays only mere possession of resources will not bring success for firms in the long run; instead, how firms use the capabilities of these resources by investing in both human and infrastructure resources to gain performance and competitive advantage.

In marketing literature, RBV has been extensively used to understand the firm's capabilities and their effect on its performance (Durmusoglu & Kawakami, 2021; Trainor, Andzulis, Rapp, & Agnihotri, 2014). Findings from the extant research suggest a positive correlation between a firm's capabilities and performances. Since the central notion of RBV explains a firm's resources and capabilities, such as operational capabilities (Yu, Ramanathan, & Nath, 2014); supply chain capabilities (Morash & Lynch, 2002), marketing analytics capabilities (Rahman et al., 2021a) and so on, therefore, we take this notion to understand how marketing analytics capability in export-oriented manufacturing firms exposes the way of success to gain sustained competitive advantage. While firms focus on internalization in the international competitive arena, it is particularly prevalent to deal with uncertainties. Firms must adapt fast decision-making processes and transferrable internal capabilities such as marketing analytics to expand the business internationally.

Further, dynamic capabilities have primarily been introduced and explored within strategic management literature. Later, other pertinent disciplines have been adopted to understand the pivotal role of dynamic capabilities in marketing, operations, innovation, and international business (Akter, Wamba, Mariani, & Hani, 2021b; Zhou & Li, 2010). Dynamic capabilities are a higher level of competencies and skills the firm possesses to create, integrate and reconfigure both internal and external resources to shape and respond to ever-changing market environments (Lee, Narula, & Hillemann, 2021). Earlier, management scholars conveyed the relation of the concept of dynamic capabilities with the RBV of the firm to explain that firm's value and profits are generated by the capabilities, assets, organizational processes, knowledge, etc., of the firm. Through these capabilities, firms implement their tactical strategies to compete (Barney, 1991). But later, scholars exhibited a clear distinction from RBV and asserted that dynamic capabilities enables firms to deal with internal and external business environments by reconfiguring, building and shaping organizational processes (Eisenhardt & Martin, 2000; Teece, Pisano, & Shuen, 1997). Scholars also argued that it is not sufficient to have specific resources and capabilities; instead, firms require dynamic capabilities of the resources (Wu, 2010).

Dynamic capabilities drive organizational innovation abilities and capacities to align resources with dynamic environmental changes, which fosters transformation with a proactive approach that differentiates and establishes a firm's decision-making as a dynamic capability in comparison to regular operational level decisions (Hughes et al., 2019; Wang & Ahmed, 2007). Teece (2007) explained and distinguished dynamic capabilities into three specific dimensions to sustain competitive advantage: (a) market sensing capacity to identify opportunities and threats; (b) market seizing capacity to grab those opportunities; and (c) enhancing and reconfiguring both intangible and tangible resources to uphold competitiveness in the marketplace. A firm's market orientation strategy consists of either a market-driving or market-driven approach. Firms leverage dynamic capabilities in a proactive-explorative while following the market-driving strategy and utilize dynamic capabilities in a reactive-exploitative way while following the market-driven strategy (Fischer, Gebauer, Gregory, Ren, & Fleisch, 2010; Randhawa et al., 2021).

Dynamic capabilities are considered a way for a firm to renew its resource base, which subsequently resolves rigidity in capabilities, followed by creating value to drive the overall organizational performance (Michailova & Zhan, 2015). Dynamic capabilities have also been studied extensively, like RBV, to understand how business firms can better use their resources and capabilities to compete in market environments. In our research context, we assume that when export-oriented manufacturing firms approach foreign buyers and big brands, they leverage marketing analytics capability to sense, seize and reconfigure the available opportunities while considering putting their footprint in the international market.

Notably, the analytics capability of the firms plays a pivotal role in the current dynamic digital context where firms have access to big data and artificial intelligence. Firms that do not have appropriate resources and associated capabilities like marketing analytics capability are unlikely to succeed in approaching the international market. In a recent study, Gupta et al. (2020) exhibited how big data predictive analytics capability drive superior organizational performance. Akter et al. (2020a) introduced service analytics capability and its impactful role across service life cycle to deliver invaluable decision-making insights, particularly in the digital marketplace. Despite the potentiality of marketing analytics capability to improve firm performance and competitiveness, untapped and unexplored opportunities are available for market researchers to study for 'deeper insights' (Cao et al., 2019). This study offers to advance our understanding using dynamic capabilities of how marketing analytics capability drive sustained competitive advantage for export-oriented manufacturing firms while broadening their business scope by approaching B2B companies in international markets.

2.2. Study urgency

Despite the significance of analytics and AI, most evidence in the academic literature consider the aspects from a B2C perspective, where only a handful of studies investigate B2B environment; although the business firms are directly dealing with the ultimate consumers (e.g., Cao et al., 2019; Davenport et al., 2020; Hossain et al., 2021). There are only a few studies in B2B and advanced analytics in industrial marketing and business management outlets (Keegan et al., 2022; Mikalef et al., 2021a). Such research scarcity is even more evident in pure B2B manufacturing firms' context. In recent times, manufacturing firms have emphasized extensively on data analytics, which is an invaluable resource for processing critical information and differentiating themselves from others (Paiola & Gebauer, 2020). Around 45% of manufacturing firms face difficulties enhancing their business due to a lack of knowledge of advanced data usage and monitoring required for viable business models to operate in a specific market (Adrodegari et al., 2015).

Further, the business operation of an export-oriented manufacturing firm is a complex system comprised of multiple functional areas that are mutually dependent on one another. In general, manufacturers can tackle the challenges of the export market barriers by utilizing advanced analytics efficiently (Bertello, Ferraris, Bresciani, & De Bernardi, 2021). However, advanced data analytics in manufacturing firms lags in penetration and diversity compared to other industries due to the lack of knowledge (Kozjek, Vrabič, Rihtaršič, Lavrač, & Butala, 2020). Moreover, a focus on the export-oriented manufacturing industry purely from the B2B and advanced analytics aspect is almost absent in the literature (e.g., Akter et al., 2021b; Baabdullah, Alalwan, Slade, Raman, & Khatatneh, 2021; Kushwaha, Kumar, & Kar, 2021). Many of the manufacturing firms in the RMG industry are currently struggling to meet business buyers' expectations due to the lack of knowledge in advanced technological orientation. (Fox, Goldrick, Peng, & Whittington, 2018; Saha, Dey, & Papagiannaki, 2021). Firms are unaware of the market-centric analytics capabilities they should possess to convince their business buyers by assessing ultimate customers' demands (Ali et al., 2021; Chaudhary et al., 2020; Lay, 2019).

The literature highlights the essential role of organizational resources and capabilities in driving a firm’s performance to gain and sustain a competitive advantage. Accordingly, we adopt a novel focus that complements and integrates with the existing literature by considering the firm’s marketing analytics capability ingrained in the theory of resource-based view (RBV) and dynamic capability. Therefore, we use the overarching theoretical lens of RBV and dynamic capability that contributes significantly to the progress of this prioritized research context. Figure 1 shows the flow chart of the study’s structure.

Insert Figure 1 here

2.3. Problem authentication

To authenticate the problem more precisely, the study considered a country—Bangladesh, which heavily relies on RMG manufacturing and export. Since the last two decades, exporting manufactured goods has become the backbone of Bangladesh's economic success.

The country is one of the world's largest apparel exporters, with the RMG manufacturing sector accounting for 84% of Bangladesh's exports[1] (Berg, Harsh, Hedrich, & Magnus, 2021). Sydney to London, London to Paris, Paris to New York, many of our clothing items hold the label "made in Bangladesh. As part of Study 1, in ***study 1a***, this research was initiated with proper problem verification from relevant news outlets. It seems, despite the country's manufacturing progress in the last two decades, the technology infrastructure remains one of the biggest challenges facing the sector (RMG-Times, 2018). This manufacturing industry is currently lagging in using advanced technology and consequently losing its market shares; thus, firms should apply analytics and AI to get ahead of the global competition (BBC-News, 2019; Samakal-News, 2019). Proper utilization of analytics and AI could gear up the industry to the next level (Hu, Nayyar, & Sharma, 2021). This industry sector needs to strengthen its digitization infrastructure (Samakal-News, 2019). Experts believe the industry is not moving fast due to a lack of knowledge and awareness of advanced technology that could adequately detect the buyers and meet the export market demand (BBC-News, 2019; Berg et al., 2021).

Further, as part of study 1, in ***study 1b***, interviews were conducted on 25 managers of manufacturing firms based on a judgemental sampling technique to verify the problem. The managers were involved with the firms that incurred losses in two consecutive years due to a lack of knowledge in data analytics. Overall, they were unsure what types of technology orientation are required and how they could reap the market success in this competitive, data-rich export business environment. For example:

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[1]A large number of RMG manufacturing firms are based in Bangladesh and they produce clothing and fashion items for almost all the world's leading brands like Walmart, New Look, Peacocks, Austin Reed, Esprit, H&M, Tesco, Aldi, John Lewis, Next, Primark, M&S, Zara, GAP, Macy, Target, K-mart, etc (Apparel-Online, 2016; Desiblitz, 2020). According to the world trade organization's data evidence, Bangladesh is one of the leading global RMG producing countries that worth US$28 billion (World-Trade-Organization, 2021).

Interviewee # 3, Male 40s: *"We are not in a position to make the right offer to the right buyer. We follow the traditional approach. When orders are received from the buyer, then we proceed. The scenario is different in some RMG firms, those that have proper data analytics capability. They can analyze all buyers and markets data, and they are capable of making the offer upfront to the buyers. Not only these, but other issues might be possible to tackle using past and current data analysis. We are not aware of all of these. It would have been great if we could get a guideline".*

Interviewee # 17, Female 50s: *"I am not aware of the specific benefit of using advanced technology. I don't think we at the firm level will be able to implement advanced analytical tools soon. We at first should know what exact benefit of using advanced technology in the RMG sector. We are incurring loss; it's true. Other firms might be doing things differently that we don't know. We don't have the evidence; exactly what they are doing differently to accelerate their business".*

In summary, interviewees highlighted the overlooked opportunities of advanced analytics that they may adopt to enhance their business performance. For example, using Amazon AWS or Microsoft Azure market-centric analytics platform, manufacturing firms may get meaningful insights from the transaction pattern of business customers. The process may help to satisfy specific demands in real-time. Moreover, all the interview participants portrayed the lack of knowledge that prevails in the manufacturing industry. This knowledge gap is deterring the advancement in the industry by jeopardizing the existence of recurring losses in this current competitive business environment.

3. Hypotheses development and model conceptualization

3.1. Marketing analytics capability

The concept of marketing analytics has emerged from the business analytics domain. Marketing analytics refers to collecting, managing, and analyzing information and data to derive valuable insights for effective marketing decision-making and gaining competitive advantage (Germann, Lilien, & Rangaswamy, 2013; Hanssens & Pauwels, 2016; Wedel & Kannan, 2016). Lilien (2011) defined marketing analytics as a technology that enables model-supported methods by using customer and market data to improve marketing decision-making. Recent studies highlighted the direct impact of marketing analytics on firm competitiveness and performance (Xu, Frankwick, & Ramirez, 2016). Marketing analytics is defined as an advanced digital method to manage and organize marketing-oriented data from a data-rich business environment to improve and support a firm's decision-making and success in today's competitive business marketplace (Wedel & Kannan, 2016). Marketing analytics plays a central role in fulfilling the demand for data and insights by developing adequate metrics and analytical methods, enabling firms to have better visibility and decision-making capability for favorable and higher business performance (Germann et al., 2013). The proliferation of digitalization encourages exploring various tools and methodologies to capture and analyze data. Marketers and researchers benefit by utilizing available resources, including web-based interactive survey tools, big data analysis through flexible programming, data mining, visualization, natural language processing etc. Marketers have access to real-time data analytics, which delivers advanced insights for buyers purchase behaviour, location, and characteristics to predict future customer behaviour that subsequently assists marketers in making faster marketing decisions and implementing the strategy (Iacobucci, Petrescu, Krishen, & Bendixen, 2019).

Marketing analytics helps to convert a large amount of unstructured data into valuable business information, which brings increased efficiency in decision-making and subsequently achieves increased revenue and decreased operational expenses for the firms (Weinberg, Milne, Andonova, & Hajjat, 2015). Therefore, marketing analytics capability is denoted as an integrative process that embraces various information sources and data modeling from multiple channels to build product/service differentiation that meets buyers' needs to ensure firm sustained performance (Song, Di Benedetto, & Nason, 2007; Wade & Hulland, 2004). A firm's marketing analytics capability integrates structured and unstructured big data to generate essential insights of suppliers, customers and competitors, which subsequently used by the decision-makers to redefine their marketing mix strategies by offering new products/services, price models and new promotional channels (Iacobucci et al., 2019; Wedel & Kannan, 2016). Notwithstanding the importance of marketing analytics, there is a lack of research studies and findings on marketing analytics capability (Rahman et al., 2021a). Scholars have also recommended investigating a firm's distinctive analytics capability instead of repeatedly exploring broader capability contexts (Hossain, Akter, & Yanamandram, 2020a). Despite the promising opportunities to apply analytics throughout the value chain, comparatively, the readymade garment business lags behind in adopting analytics, often prioritizing "gut feel"-driven decision-making by merchants and designers (Fox et al., 2018). However, some firms have already cracked the beauty of marketing analytics and adopted it in their business operations. For example, Beximco, a leading manufacturing company in Bangladesh and partner of major brands like Zara, U.S. Polo Assn, Marks & spencer, etc., is using market-centric analytics with their own R&D for forecasting, analyzing trends, and designing fabrics for the mentioned brands by taking into consideration their presence in the foreign markets (Forbes, 2021). Interestingly, Raymond, another RMG brand and the leading supplier in the apparel industry in India, has adopted AI-driven market-centric analytics capability. The system implements data analytics-driven sales automation to connect RMG retailers and dealers across the country with whom Raymond used to do coordination manually at an earlier stage (Ahaskar, 2019).

Our study aims to extend the current knowledge of marketing analytics capability in the context of export-oriented manufacturing firms. We assume that when these firms aspire to expand the business and approach foreign buyers, marketing analytics capability contributes significantly to understanding the respective market, customers' trends, purchase behaviour, competitors, opportunities, etc. These manufacturing firms utilize their marketing analytics capabilities by considering all the available data from the international market to design innovative products for the buyers/customers in the international markets and multinational brands. While considering foreign market buyers, manufacturing firms are the supplier for that market and compete with other local and international suppliers to capture the market opportunities. Therefore, it is deemed that export-oriented manufacturing firms with distinctive marketing analytics capabilities can differentiate from the competitors to achieve superior business performance and sustained competitive advantage.

3.2. Market sensing

Sensing is considered an indispensable part of dynamic capability and business strategy (Teece, 2014). Market sensing capability portrays the firm's abilities to continuously scan, search and explore across technologies and markets for opportunities (Teece, 2007). So, it specifies organizational requirements for assessing market/buyer's needs, resource integration and technological possibilities/capabilities to identify new market opportunities. The firm's internal capabilities make it possible to recognize opportunities in the market by interpreting, analyzing, and utilizing this invaluable information (Heusinkveld, Benders, & van den Berg, 2009). Firms need to develop the right resources and use the assistance of relevant instruments to recognize market changes, which is a top strategic priority for firms (Endres et al., 2020). Market sensing capabilities promote product innovation and drive speedy market launches by improving new processes and product development effectiveness and efficiency. Market sensing capabilities act as a wider version of an organization's market intelligence to deal with buyers current and future demands, which are accommodated by disseminating knowledge across functional units (Jaworski & Kohli, 1993). Firms with good sensing capabilities have a high level of alertness to detect both opportunities and threats. Business firms can establish a link with the target market through their market sensing capabilities. In summary, market sensing capability is the ability of a firm to collect, analyse and filter market information obtained through market intelligence from internal and external resources to understand the importance of market information, followed by determining implications for actions to increase opportunities and reduce uncertainties for introducing innovative solutions for the market and buyers (Day, 1994; Lin & Wang, 2015). Firms can achieve a superior value proposition by methodologically adapting operations and processes to synchronize with market changes (Bharadwaj & Dong, 2014). Export-oriented manufacturing firms focusing on business expansion in a foreign market require market sensing capabilities to identify possible opportunities and threats. Since uncertainties are inevitable while starting a business in an international market, these manufacturing firms are expected to leverage market sensing capabilities to learn about buyers, consumers, rivals, channel partners, and the overall market environment.

3.3. Market seizing

Firms identify both opportunities and threats as a part of their business operations. Firms make a pivotal decision on how to respond to market changes and whether/how to grab pertinent opportunities for firms to achieve business expansion and growth. To enhance the current state of value propositions, firms need to explore and validate a wide range of opportunities (Teece, 2007). Market seizing generally emphasizes responding to newly identified opportunities and threats. Seizing refers to a firm's capabilities of deploying resources by mobilizing both internal and external resources that enable firms to make decisions for required changes to deal with opportunities and threats (Teece, 2014). Firms, who have the ambition to achieve sustainable innovation through their products and services, commensurately market seizing capabilities contribute significantly towards accomplishing the ambition (Mousavi & Bossink, 2017). Firms may have capabilities to sense market opportunities, but firms must have substantial managerial capabilities to seize the opportunities identified through market sensing.

Business firms need proper strategic planning to exploit new opportunities and appropriate investment decisions and business models, developing technological competencies followed by acquiring both tangible and intangible assets (Khan, Daddi, & Iraldo, 2021; Teece, 2007). Once business firms scan and test for new business ideas, they need to adjust their decisions and business models to materialize them. Therefore, the firms enter into the seizing stage by possessing the capabilities with the help of technology, resources and market expertise (Inigo, Albareda, & Ritala, 2017). It highlights the essentiality of establishing and maintaining connections with local ecosystems and analyzing the market facts and B2B context (Lütjen et al., 2019). Market seizing also emphasizes continuous commitment to research and development (Mikalef et al., 2021a), through which firms can mobilize their resources for both sensing and seizing opportunities. With the help of analytics capability, business firms can better sense the opportunities, which indeed help them to seize market opportunities by avoiding probable threats to cope with the incessant changing environments (Alinaghian, Kim, & Srai, 2020). In a B2B environment, manufacturing firms require market seizing capabilities to address market needs that drive business expansion in a new market environment. Therefore, export-oriented manufacturing firms will likely set targets by focusing on strategic planning and resource development to seize market opportunities.

3.4. Market reconfiguration

Firms need to reconfigure their resources to prepare for responding to market changes. Market reconfiguring capabilities refer to the continued renewal of existing resources to reconfigure fundamental capabilities (Teece, 2014). Firms require to adapt to market changes by reconfiguring the extent to which firms are currently conducting their business operations (Teece, 2007). Firms put an extensive effort at the reconfiguring stage by developing and restructuring new organizational culture and work processes for change management (De Silva, Al-Tabbaa, & Khan, 2021; Teece, 2018). A firm's continuous improvement processes help build required competencies over time that redirect towards enhancing the firm's market reconfiguration capabilities for long-term sustained performances (Wilden, Gudergan, & Lings, 2019). Once firms sense and seize market opportunities, they must combine their resources to acquire the opportunity through market reconfiguration capabilities. Firms with limited capabilities for sensing and reconfiguring may face organizational inertia that increases operating costs and minimizes growth for the business (Helfat et al., 2009). Since the business environment is dynamic, firms need to either follow market-driving or market-driven strategies. For instance, export-oriented manufacturing firms in a B2B environment with a market driving focus can sense opportunities ahead of competitors. Frequent reconfigurations of resources enable them to offer new solutions (Wilden et al., 2019). On the other hand, market-driven firms collect information from the market, and they highly rely on real-time data followed by a strong communication network with external market, which requires strong sensing capabilities to go for a phase-wise approach for resource allocation and reconfiguration (Slater & Narver, 2000; Wilden et al., 2019). Therefore, irrespective of adopting a market-driving or market-driven approach, export-oriented manufacturing firms must comply with adequate sensing and reconfiguring capabilities to seize market opportunities. Dong (2019) articulated how seizing new entrepreneurial opportunities by continuous digital transplantation and reconfiguring of digital platforms facilitates firms to expand business from one sector to another. Moreover, firms may also need to reconfigure their existing business model for new industry or market entrance (Battistella, De Toni, De Zan, & Pessot, 2017). A firm's resilience is reinforced with the help of resources reconfiguration. Thereby, export-oriented manufacturing sectors require the agility of reconfiguration capabilities to become successful in business operation and expansion.

Organizations leverage on their resources and internal-external capabilities like analytics capability to develop and enhance market sensing, seizing, and reconfiguring capabilities. A firm's dynamic capabilities such as sensing, seizing, and reconfiguring can be enhanced through creating distinctive business value such as business analytics, which works as an enabler for achieving dynamic capabilities (Conboy, Mikalef, Dennehy, & Krogstie, 2020; Mikalef, van de Wetering, & Krogstie, 2021b). Analytics capabilities help firms develop dynamic capabilities to compete for challenges and take advantage of opportunities available in the marketplace. Akter et al. (2020a) articulated that service system analytics capabilities positively impact the organization's market sensing, seizing, and reconfiguring capabilities. In line with the discussion, we also argue that marketing analytics capabilities adopted by export-oriented manufacturing firms will act as an enabler to develop dynamic capabilities (market sensing, seizing and reconfiguration capabilities). Thus, we put forward the following hypotheses:

H1a–H1c. Marketing analytics capability positively influences (a) market sensing, (b) market seizing, and (c) market reconfiguration.

Competitive advantage gives a business the leverage it needs to differentiate itself from competitors by exceeding buyers' expectations with value. Competencies and internal intellectual assets reflect a knowledge-based perspective and can assist businesses in developing superior goods and services that provide a long-term sustainable competitive advantage (Bharadwaj, Varadarajan, & Fahy, 1993). Dynamic capability theory portrays more internal organizational capabilities (e.g., marketing analytics capability) rather than external capabilities to enhance the firm's competitive advantage (Wu, 2010). The firm's market sensing, seizing and reconfiguring capabilities are the fundamental premises of dynamic capability to adopt flexibility in business operation that enable firms to compete in a rapidly changing environment (Ambrosini, Bowman, & Collier, 2009). Business firms achieve the first-mover advantage with their excellent market sensing, seizing, and reconfiguring capabilities by ensuring reduced response time, improved delivery performance and a higher level of customization (Akter et al., 2020a). A firm's market sensing, seizing, and reconfiguring capabilities are pivotal in building a competitive advantage. Market sensing capability helps to combine information with the help of analytics to take strategic decisions for implementing innovative solutions to meet buyers' needs and eventually enhance competitive advantage. Market seizing capability alleviates the firm's performance to achieve competitive advantage by addressing market opportunities by mobilizing their resources. Market reconfiguring capability enables firms to transform their business operations according to changing market demand by enhancing flexibility and achieving sustained competitive advantage. Thereby, we hypothesize that:

H2a–H2c. (a) Market sensing, (b) market seizing, and (c) market reconfiguring abilities are positively associated with a firm's sustained competitive advantage.

A firm's analytic capability drives and facilitates dynamic capabilities to enable firms to respond to market changes and subsequently achieve competitive advantage (Erevelles, Fukawa, & Swayne, 2016). Analytics such as marketing analytics capability exposes opportunities for firms with insights required for strategic market planning that eventually enables firms to build market sensing, seizing and reconfiguring capabilities, hence forming dynamic capabilities (Fischer et al., 2010). Dynamic capabilities are indispensable for firms to achieve competitive advantage by managing resources with reduced response time and higher delivery performance. Commensurately, marketing analytics capabilities provide necessary resources (e.g., insights, intelligence) to enhance their dynamic capabilities (sensing, seizing, and reconfiguring) to achieve a competitive advantage in the challenging market environment. As a corollary to the above-mentioned discussion, extant research also highlights that dynamic capabilities act as critical mediators to create a competitive advantage for business firms (Cao et al., 2019; Hsu & Wang, 2012). In a recent study, Akter et al. (2020a) empirically demonstrated that dynamic capabilities (market sensing, seizing and reconfiguring) have a mediating relationship between service systems analytics capabilities and competitive advantage. Therefore, in line with this research's theoretical perspective and prior study findings, we contemplate that marketing analytics capability improves a firm's dynamic capabilities, leading to sustained competitive advantage in export-oriented manufacturing firms' context.

H3a–H3c. The relationship of marketing analytics capability and sustained competitive advantage will be mediated by (a) market sensing, (b) market seizing, and (c) market reconfiguration.

3.5. Adoption of artificial intelligence

The proliferation of AI in operational settings has gained considerable attention from practitioners. Industrial marketing research suggests that "AI technologies facilitate controlled and dynamic learning" (Agnihotri, 2020, p. 298) and its adoption can enable business firms to mitigate challenges in competitive business environments and improve firm performance (Baabdullah et al., 2021). AI is frequently characterized as "a system's ability to interpret external data correctly, to learn from such data, and to use those learnings to achieve specific goals and tasks through flexible adaptation" (Kaplan & Haenlein, 2019, p. 17). Despite the formal definition, AI has versatile breadth and depth of applicability in practice and research (Stahl et al., 2021). Organizations use data and analytics to make better and more informed decisions in the competitive business environment. More business firms adopt a data-driven approach to create flexible and accurate decisions based on data and analytics, powered by AI as the backbone. Therefore, firms are investing in developing information technology (IT) capabilities like AI to maximise operational excellence and return on investment (Rahman et al., 2021a).

The broader perspective of IT-business research articulates that leveraging on IT and AI organizations develops their dynamic capabilities, which leads to achieving competitive advantage for the firms (Benitez, Castillo, Llorens, & Braojos, 2018). In a B2B environment, external market and customer knowledge are generated by embracing big data-driven AI, which also help firms to take B2B marketing rational decisions to enhance performance (Bag, Gupta, Kumar, & Sivarajah, 2021). This recent work highlighted the power of AI as a driver for automation of data and analytics and influential for B2B marketing knowledge management processes. In a recent study, Mikalef et al. (2021a) elaborated the application of AI with case studies on how AI can be leveraged for B2B business operations from a B2B marketing perspective. A 15 to 20 percent more productivity in the RMG industry is possible by utilizing AI; China, a developed country, has already started using AI extensively to achieve enhanced production and sustainable export growth (Amin, 2022). Adopting AI is expected to bring various benefits to the manufacturing industry, such as improving material grading, automating data-gathering with asset management, minimizing errors in the final product inspection, and so on (Partida, 2022). A pioneer B2B sports brand, Tianyuan Garment, a major supplier for Adidas and Reebok, has introduced AI-driven robots that take roughly four minutes to cut the cloth and make a T-shirt, reducing the production cost for each unit by 33% (Barrie, 2019). A Korean textile firm, Hyosung TNC has also implemented AI in manufacturing systems that capture and analyze data from the whole supply chain from raw material import to final product export, delivering consistent quality products to business buyers (Jin & Shin, 2021). Evolutionary AI enables RMG firms to cover a wide variety of operations, including but not limited to: design aids, fashion suggestions based on user feedback, intelligent tracking systems, textile quality assurance, trend foresight, and supply chain decision making (Thomassey & Zeng, 2018).

AI is an emerging technology, and because of its evolving dynamism, scholars highlight the need for studies within the B2B industry context, examining how the adoption of AI impacts business capabilities development and deployment (e.g., Agnihotri, 2020). Notwithstanding the substantial contribution of AI to a firm's performance, there is a lack of understanding and debate on how AI can play a role in automating data and applications for B2B firms to generate invaluable insights about the market and buyers/consumers (Mikalef et al., 2021a). A leading B2B fashion manufacturing company in India, Fashinza introduced a manufacturing platform powered by AI and data science algorithms to deliver a one-stop ‘marketplace’ for fashion brands and retailers by maintaining a database of garment histories, which is then used to educate algorithms that pair brands with appropriate suppliers and predict metrics like turnaround time (Wiggers, 2022). In Bangladesh, brands like Pacific Jeans and Beximco, major suppliers for top retail brands like Zara, Gap Inc, Nike, etc., have started using ERP systems powered by advanced artificial intelligence to decrease fabric waste and streamline manufacturing operations (PiStrategy, 2021). We take this notion to understand how export-oriented manufacturing firms, particularly, can leverage the adoption of AI to develop dynamic capabilities with the help of marketing analytics. Business performance driven by data analytics capabilities and strategic decision-making varies with the adoption of a higher level of technologies like AI (Fethi & Pasiouras, 2010; Rahman et al., 2021a). Moreover, business firms that adopt AI possess distinctive competencies compared to no adopters, therefore driving better control of their business operations, marketing planning and implementation, and internal and external resource allocation (Martínez-López & Casillas, 2013). In a recent study, Rahman et al. (2021a) empirically exhibited that the adoption of AI strengthens the connection of marketing analytics capability on competitive marketing performance. We argued that marketing analytics capability helps building and enhancing market sensing, market seizing and market reconfiguration capabilities. As a corollary of this previous point, we posit that the adoption of AI by export-oriented manufacturing firms strengthens the impact of marketing analytics capability on enhancing market sensing, seizing and reconfiguring capabilities of firms. Accordingly, this study suggests the following hypotheses for further investigation:

H4a–H4c. The adoption of artificial intelligence moderates the relationship between marketing analytics capability and (a) market sensing (b) market seizing and (c) market reconfiguration.

3.6. Conceptual model

In ***study 2a***, the literature review helped hypothesize the relationship between marketing analytics capability, artificial intelligence, market sensing, seizing, reconfiguring, and competitive advantage (see previous sections). Further, two Delphi studies were conducted to conceptualize the research model. In ***study 2b***, interviews were conducted on 25 export-oriented manufacturing firms managers who were actively involved in handling business customers' data. This process of sampling was judgemental. It seems all the managers acknowledged the importance of market-centric analytics capacity and AI in order to sense, seize, reconfigure the offers and market-centric activities. The analytics and AI-driven process provide them the ample opportunity to enhance sales growth and profitability. For example:

Interviewee # 7, Male 40s: *"Data-driven strategy-wise, market success is straightforward in my view. We not only possess buyers' and suppliers' information but also analyze consumers' data. We have market-centric analytics advancement that can even* *capture google searching data from a specific region. Based on the evidence, we are capable of approaching buyers first. It works like magic. Most buyers become impressed by looking at our sensing and apprehending capacity. In this way, we are penetrating our foreign market".*

Interviewee # 19, Male 40s: "*As an analytics manager of company X, I have closely seen our business achievement. The central myth behind our sustained revenue is data-driven innovation. We are one step ahead of our competitors. If any alteration is required in our marketing process, we have advanced market-centric analytics support to reconfigure the process. We rely on analytics from product design to offer and use the AI-driven process to make things more accurate. For example, based on market-centric analytics capability, we offer our valued buyers rather than wait for their order. Further, our AI adoption helps product design and provides some real-time solutions to our buyers".*

Hence, manufacturing firms with a focus to expand their business in the foreign market are leveraging on both marketing analytics capabilities and AI-driven process implementation to understand the ongoing trends to deliver context specific solutions for international buyers. Importantly, firms commensurately identify opportunities that enable to offer a comprehensive package for foreign RMG buyers scratching from new design development to implementation. Moreover, all the interviewees explicitly emphasized on adoption of marketing analytics and AI to achieve enhanced capabilities for sensing, seizing, and reconfiguring to creates noticeable difference from the competitors, which indeed drives firm's overall performance and sustained competitive advantage. Furthermore, a text analysis through a word cloud (figure 2) is depicted below to better understand the qualitative synthesis extracted from the interview transcripts. Text analysis helps to capture un-structured or semi-structured data with an aim of summarizing considerable amount of textual information for further critical exploration and reflections (Aggarwal & Zhai, 2012). The entire text cloud (figure 2) implied that participants highlighted similar concepts related to data-driven marketing analytics, capability, AI, and related benefits of sensing, seizing, and reconfiguration.

Insert Figure 2 here

Besides, following high-impact research guidelines (e.g., De Luca, Herhausen, Troilo, & Rossi, 2020), in ***study 2c***, we forwarded a survey link to 135 export-oriented RMG manufacturing firms managers using snowballing sampling techniques. We received 39 responses (28.9% response rate). 89% of all items were broadly overlapped with our study constructs. This evidence confirms our study constructs' parsimony and relevance in the context of export-oriented manufacturing firms B2B aspect in a data-rich environment. Following the guideline from studies 2a, 2b, and 2c, the study proposes the following conceptual model (figure 3) for empirical investigation in the context of B2B export-oriented manufacturing firms.

Insert Figure 3 here

4. Model validation: methods and analysis

4.1. Measurement scales

In ***study 3a***, an extensive context-wise review was conducted, and all instruments of the research model constructs were adapted from prior studies. For example, market analytics capability instruments were adapted from Rahman et al. (2021a). Market sensing, seizing, and reconfiguring instruments were adapted from Akter et al. (2020a), adoption of artificial intelligence items were taken from Rahman et al. (2021a), sustained competitive advantage items were taken from Cao et al. (2019). In ***study 3b***, the authors followed the guideline of Rahman, Hossain, Fattah, and Mokter (2021b) and Ou, Verhoef, and Wiesel (2017) to make sure the quality of the instruments in this study context. Overall, the process helped to ensure the face validity of the items and reduce the items to enhance the response rate. In this phase, using the judgmental sampling technique, researchers invited 15 managers and 15 academic scholars knowledgeable in this study area. In this phase, the survey instruments were checked deeply and corrected the items based on the experts' opinions. In ***study 3c***, pilot surveys were conducted on 75 representative samples using simple random sampling technique. Statistical analysis of the respondents' responses confirmed the measurement scales' reliability, validity, and overall functionality. This research's all qualitative and quantitative phases were initiated after taking appropriate ethics approval.

4.2. Main survey, sampling and demographic profile

In ***study 4a***, the authors recruited one of the top research firms to collect data from managers of export-oriented RMG manufacturing firms. Qualtrics version of the questionnaire link was distributed via the research firm's database using the random sampling technique. In total, 1953 managers were approached. The questionnaire contained the screening questions that screened the managers who have knowledge of using advanced technology in their business operations in order to penetrate the export markets. 1675 managers were screened out in this process, and 278 managers were qualified for the final steps. Further, the authors checked the straight liners, attention check questions, and total response time. After investigating the odd cases, the authors finally considered 257 responses for data analysis. The demographic data shows 56.4% of managers are male, 42.4% female, and 1.2% are not interested in disclosing their identity. 54.9% of respondents' firms are small and medium, whereas 45.1% are large. 82.4% of them have more than three years of data analytics experience (See Table 1).

Insert Table 1 here

4.3. Common method variance (CMV)

The study concentrated on Hulland, Baumgartner, and Smith (2018) guidelines to tackle the common method bias in survey research. Following the guideline, this study applied both priori and post hoc methods. As part of the priori method, survey questionnaires items of each variable (e.g., Independent, dependent, mediator, and moderator) were placed in a different section. The study was applied the different response scales for independent and dependent variables. Respondents' identities were kept confidential. Besides, paired t-test on two random split sets of data confirmed there was no issue of non-response bias. The study further considered the post-hoc method. The study had two questions as part of the marker variable. The co-relation of marker variable and study constructs variables are low and insignificant justify the study constructs correctness (r = - .319 to .011, where, p>0.05).

4.4. Addressing unobserved heterogeneity

Apart from the observed variables, some unobserved variables could affect a firm sustained competitive advantage. For instance, some businesses might hold strong integrity in data analytics than others, or employees of some firms might have more skill in handling analytics platforms. In this study, researchers were not able to observe those facts. Not considering such unobserved facts may lead to uneven parameter estimates and statistically biased effects. To tackle unobserved heterogeneity issues, the study used a semiparametric approach following De Luca et al. (2020)'s procedure. It depicted the intercept stint with a finite number of approval points to consider for heterogeneity of entire firms.

4.5. Addressing endogeneity issue

Further concerns regarding endogeneity may occur because some omitted variables might influence both perceived sustained competitive advantage and crucial illustrative variables. Firstly, an enterprise's strategic intention to build in data analytics is constructed in expectation of future performance (Yasmin, Tatoglu, Kilic, Zaim, & Delen, 2020); thus, marketing analytics capability could be endogenous. Secondly, all three dynamic capability outcome factors (sensing, seizing, and reconfiguring) rely on both firm policies and the capacity to analyze available data (Akter et al., 2020a), and the latter are unobservable to researchers. Thirdly, competitive advantages not only rely on firms' capability but also depend on competitors' actions and market acceptance (De Luca et al., 2020). Such external influences were unobservable to researchers. As locating proper tools was not viable, the study followed Ebbes, Wedel, Böckenholt, and Steerneman (2005) approach to apply the latent instrumental variable.

4.6. Measurement model

The PLS algorithm verifies the reliability of the model construction. The loadings on all structural elements exceed the critical value of 0.70, ensuring the reliability of the measurement items. Further composite reliability (CR) of constructs are above 0.80 prove adequate. The average variance extracted (AVE) of the variable values exceeds the threshold of 0.50 (Hair, Risher, Sarstedt, & Ringle, 2019). Detailed results are shown in Table 2. VIF values range from 1.342 to 2.853, meeting criterion ≤3. All these configurations support convergent validity. Results also support discriminant validity. Results from both the Fornel-laker test and the heterotrait-monotrait ratio (HTMT) are adequate. HTMT values are below 0.85 (Henseler, Ringle, & Sarstedt, 2015), and the results in Table 3 show that all correlations between constructs are lower than the square root of AVE (Hossain et al., 2021), confirming the discriminant validity of the study.

Insert Table 2 here

Insert Table 3 here

4.7. Structural Model

The results show the hypothetical connections in a nonparametric way that used 5000 subsamples in bootstrap. The connections of H1a: MAC-MSN (β=0.657, p<0.001), H1b: MAC-MSI (β=0.662, p<0.001), H1c: MAC-MRN (β=0.691, p<0.001), H2a: MSN-SCA (β=0.370, p<0.001), H2b: MSI-SCA (β=0.242, p<0.001), and H2c: MRN-SCA (β=0.216, p<0.001) are significant. The results confirm H1a-c and H2a-c hypothetical relationships. Following Preacher and Hayes (2008) and Hayes, Preacher, and Myers (2011) procedure, the mediator analysis of MAC-MSN-SCA ($\beta=0.243$, $p<0.001$), MAC-MSI-SCA ($\beta=0.160$, $p<0.01$), and MAC-MRN-SCA ($\beta=0.150$, $p<0.01$) confirms the significant effect. Thus, H3a-c hypothetical relationships are supported. Utilizing two stages process, moderator findings show AI adoption moderates the relationship of MAC-MSN ($\beta=0.104$, $p<0.05$), MAC-MSI ($\beta=0.131$, $p<0.05$), and MAC-MRN ($\beta=0.128$, $p<0.05$). Thus, H4a-c hypothetical relationships are supported (See Table 4). The MAC generates 48.2% power over MSN (R^2 = 0.482), 50.5% power over MSI (R^2 = 0.505), 53.5% power over MRN (R^2 = 0.535). Overall the model effect generate 56.8% power towards SCA (R^2 = 0.568) (See Figure 4).

Insert Table 4 here

Insert Figure 4 here

4.8. Robustness tests

The study controlled Firm size (FSI) (β= -0.001, $p>0.05$), Analytics experience (ANE) (β= 0.005, $p>0.05$), and Respondent position (RPS) ($\beta=0.075$, $p>0.05$). These control variables do not have any significant effect on the study findings. As per the guideline of the goodness of fit MAC-MSN, MAC-MSI, MAC-MRN effect sizes (f^2) are very strong (> 0.350) (Hossain et al., 2021). Using blindfolding predictive validity confirms Stone-Geisser's Q^2 values of 0.350 for MRN, 0.294 for MSI, 0.337 for MSN, 0.421 for SCA (Chin, 2010). PlS-predict approach confirms the nomological validity where 10 with 10 repetitions used calculating RMSE and residuals histograms (Shmueli et al., 2019). Model superiority over a naive prediction was confirmed as Q^2_predict values are above 0. Overall, the original model produces a smaller BIC value than the modified model where MRN was dropped, confirming the original model's genuine acceptance. Following De Luca et al. (2020) and Hulland et al. (2018) guideline, as part of study 5a, the study attempted to collect close to objective data after a certain time. Since study 4 measured both perception-based marketing analytics capabilities and sustainable competitive advantage (performance) data, additional performance data were collected to test the lagged effects of marketing analytics capability. In ***study 5a***, we were able to obtain data from a subsample of 93 participants three months after study 4 (response rate 31.3%). Perceived performance measures of time-delayed marketing analytics capability strongly correlated with sustained competitive advantage. For example, we co-related the items “Using market-centric analytics capacity, how well has your firm performed its market share growth goals” with “the percentage of market share growth” generates a strong positive correlation (r = .67). These all the procedures confirm the robustness of the research model.

Insert Figure 5 here

5. Findings and discussions

The study shows that marketing analytics capability can improve an export-oriented manufacturing firm’s competitiveness by sensing, seizing, and reconfiguring the market. Comprehending the factors that enable such possibilities and the instruments for utilizing marketing analytics capability in the manufacturing firms B2B context is critical to both enterprise and academic research. The link between marketing analytics or business analytics and a firm's competitiveness has been suggested to be very complicated (Tan, Guo, Cahalane, & Cheng, 2016). However, there is a shortage of conceptual and empirical evidence in the context, primarily focusing on the export-oriented B2B RMG manufacturing firm’s aspect (e.g., Kumar & Sharma, 2017; Rahman et al., 2021a; Sarkar & De Bruyn, 2021; Wedel & Kannan, 2016). Similarly, a study by Cao et al. (2019) focused on marketing analytics on the B2C aspect, as data evidence exposed from retail and professional services.

Therefore, based on the RBV and dynamic capabilities perspective, this study examined the following possibilities in export-oriented manufacturing firms B2B context: (a) the effect of export-oriented manufacturing firms marketing analytics capability generates market sensing, seizing, and reconfiguring ability (H1a-c), (b) the impact of market sensing, seizing, and reconfiguring enhance sustained competitive advantage (H2a-c), (c) The mediating role of market sensing, seizing and reconfiguring plays a vital role in between the relationship of marketing analytics capability and sustained competitive advantage (H3a-c). (d) The moderating effect of artificial intelligence adoption impacts marketing analytics capability and market sensing, seizing, reconfiguring (H4a-c).

The research shows that export-oriented manufacturing firms' marketing analytics capability in the B2B aspect significantly influences market sensing, seizing, and reconfiguring issues and ensures a sustained competitive advantage. The effects become much higher in the presence of AI adoption. It seems, initially, without the moderating effect, marketing analytics capability generates 46.9% power on market sensing, 48.5% on market seizing, and 51.6% on market reconfiguring. However, in AI adoption, the marketing analytics capability effects on market sensing become 48.2%, seizing become 50.5%, and reconfiguring become 53.5%. Further, increasing AI-adoption by one standard deviation unit will moderate MAC-MSN relationship by 1.983, MAC-MSI relationship by 2.533, and MAC-MRN relationship by 2.576. Figure 5 shows that even in low AI adoption and low level of marketing analytics capability, a firm can sense, seize and reconfigure the market at a slow pace, but the pace gradually accelerates when a firm utilizes high marketing analytics capability along with high AI adoption. The findings are rarely explored in contemporary studies that acknowledge the importance of marketing analytics and firm competitiveness (e.g., Davenport et al., 2020; Iacobucci et al., 2019; Rahman et al., 2021a; Xu et al., 2016). This research establishes data as the basis of marketing analytics capability and AI that helps firms sense, seize, and reconfigure the market to gain sustained competitive advantage.

5.1. Theoretical contributions

This research presents several contributions to improving the theoretical knowledge of terms related to marketing analytics capability and its effects in the context of RBV and dynamic capabilities.

First, this study combines RBV and dynamic capability view like other studies as a theoretical base; however, it extends the theory by showing how AI contributes to a firm's capability and performance, specifically in an export-oriented manufacturing firm context. Our scholarly inquiry is in line with the call for research, where scholars ask to assess what types of industries may benefit from AI usage and what would be the appropriate mechanism to adopt AI technologies (Agnihotri, 2021). Moreover, this research contributes to the analytics and AI literature by exacerbating our knowledge of the complex relations between marketing analytics capability and AI to attain distinct performance. Extant studies just concentrated on analytics capability and its direct impact on performance (Mikalef et al., 2017; Mikalef et al., 2019; Wamba et al., 2017) and AI capability and its immediate influence on performance (e.g., Davenport et al., 2020; Farrokhi et al., 2020; Fethi & Pasiouras, 2010). There is a literature gap in how analytics capability can be accelerated using AI to sense, seize and reconfigure the market. The moderating impact of AI adoption conveys the most impressive routes about how the paradigm of Analytics and AI research can progress and move forward together.

Second, this research contributes to the growth of the business/marketing/customer analytics literature. Scholars (e.g., Aydiner, Tatoglu, Bayraktar, Zaim, & Delen, 2019; Hossain et al., 2021; Rahman et al., 2021a) suggest that the connection between analytics capability and its influence on firm competitiveness/performance is a complicated method, but this research suggests and empirically test a research model linking the power of marketing analytics capability to sensing, seizing and reconfiguring the B2B export-oriented marketplace. Therefore, this study improves our knowledge of the impact of the prerequisites required to use insufficiently studied business intelligence (Bordeleau, Mosconi, & de Santa-Eulalia, 2020; Brynjolfsson & Mcafee, 2017). Extant research has concentrated on comprehending the direct impact of marketing/business analytics on firm performance/competitiveness (e.g., Aydiner et al., 2019; Iacobucci et al., 2019; Rahman et al., 2021a), but this study shows the indirect effect where marketing analytics capability first sense, seize and reconfigure the market that helps to attain the sustained competitive advantage.

Third, this study incorporates a marketing analytics capability (MAC) study with a dynamic opportunity perspective to improve our knowledge of MAC mechanisms and the effects (sensing, seizing, and reconfiguring) that can secure a competitive advantage. Although extant researches has exposed that marketing and customer analytics capacities are crucial to achieving competitive advantage (Hossain et al., 2020b, 2021; Rahman et al., 2021a), little is understood about how this can be attained in practice, especially in export-oriented manufacturing firms B2B context (e.g., Cai et al., 2021; Chaudhary et al., 2020; Colombi et al., 2018; Fox et al., 2018). This research is a pioneer in uncovering empirical proof that marketing analytics capabilities can sense, seize, and reconfigure markets and provide export-oriented manufacturing firms with a sustainable B2B competitive advantage. This research contributes significantly to dynamic marketing capability's under-examined research (Cao et al., 2019; Endres et al., 2020) in specific and on the dynamic capabilities aspect (Gupta et al., 2020; Wang & Kim, 2017) in general. Altogether, the above theoretical contributions shed new light and improve our understanding of existing export-oriented B2B marketing research.

5.2. Managerial implications

5.2.1. Transform efforts to marketing analytics actions

Managers need to be aware that marketing data is rewarded not by its quantity, speed, or diversity, but by its organizational ability to realize the potential for action provided by marketing analytics capabilities. This research introduces marketing analytics capability: a firm's ability in exclusive data analytics to understand buyers'/customers' changing demands and customize data-driven offerings based on buyers' desires. The firm should keep all the stakeholders' and market data (e.g., buyers, suppliers, competitors, consumers) in the data warehouse. The analytics will analyze the relevant data to match the offering according to prospective buyers' desires. The ability to meet buyers' expectations generally creates a distinct position in the market. Under the marketing analytics capability paradigm, a firm should collaborate with the engineering department to introduce various marketing mix models in their computer system to develop marketing plans and offerings based on available organizational resources. Protecting all stakeholders' data will also be considered a unique capability as malfunctioned data can destroy the business and business reputation.

5.2.2. Sync AI with marketing analytics strategies

Our research findings indicate the contingent development of the export-oriented RMG manufacturing industry's further digitalization using AI. It seems if an industry is at an early stage of digitalization, managers should focus on distinct marketing analytics capability. Moreover, technologies are evolving sharply, where digital data will connect 75% of the world's population by 2025 (Coughlin, 2018; Seagate, 2021). Therefore, AI's presence will significantly improve a firm's day-to-day activities. AI adoption will provide a more real-time solution. The adoption of AI in marketing analytics will accelerate firm performance in sensing, seizing, and reconfiguring the market. AI adoption on the base of marketing analytics help firm to design product and services distinctly, manage and improve supply chains and lessen shipping expenses and transit time, make more perfect predictions of inventory demand, reduce wastage, and accelerate the strategic decision process measuring marketing predictions accurately.

5.2.3. Anticipate sustained competitive advantage pathway

The research findings demonstrate that this three market-centric dynamism of sensing, seizing, and reconfiguring abilities generated through marketing analytics capability envisage the pathway towards sustained competitive advantage. Managers who wish to pursue this path should concentrate on marketing analytics capability and AI to enhance the sensing ability to detect competitors' strategies and tactics, learn both micro and macro market environments, and identify and understand the market trend. Marketing analytics capability and AI's appropriate implications further equip managers for seizing the market, finding a buyer-centric solution, adopting best practices to capture the market, responding to any market imperfections, and capturing every single opportunity from the market. The market-centric analytics capacity and AI further guide managers to reconfigure the relevant marketing processes, constantly achieve targets, and implement new marketing activities methods. These initiatives will anticipate enhancing market share growth, sales growth, profitability, and good return on investment to attain a sustained competitive advantage.

5.2.4. Connect SDG 9 utilizing marketing analytics capability infrastructure

Sustainable development goal (SDG 9) "industry, innovation, and infrastructure" is one of the 17 SDGs that has been proposed by the United Nations in 2015. World's industries have provided great attention on this aspect. SDG 9 aims to create resilient infrastructure, advance sustainable industrialization, and accelerate innovation. SDG 9 strives to accomplish targets by facilitating sustainable infrastructure improvement, particularly for developing countries, and supporting technology development and universal access to information communications technology. Our research shows how B2B export-oriented manufacturing sector managers can design data-analytics-driven infrastructure utilizing marketing analytics and AI to reap the sustained outcome in the data-rich global competitive business environment.

5.3 Limitations and future research directions

The study particularly on export-oriented RMG manufacturing industry's B2B aspect as the majority of firms are struggling, which is why the study aimed to guide them. Thus, future research may use and test the model for another industry context to re-visit the impacts. The study analyzed self-reported managers' data and matched responses with some close to factual information in the robustness test. Future studies may attempt to collect more objective data in analyzing the research findings. The final nature of the study is cross-sectional. Thus, future research may introduce the methods of longitudinal or experimental in this context. The study developed on specific variables and instruments of export-oriented manufacturing firm context. Future research should also identify marketing analytics capability and AI performance metrics in different B2B industry contexts (see table 5).

Insert Table 5 here

6. Conclusion

This research is the first empirical evidence on marketing analytics, dynamic capability, and AI from the export-oriented manufacturing industry's B2B competitive advantage perspective. Data were collected from only those managers who extensively use analytics in their day-to-day business operations, and they are well aware of the advanced technological impact. It seems those export-oriented manufacturing firms are involved in B2B marketing analytics and AI can sense, seize, and reconfigure the market. The firms are gaining a sustained competitive advantage utilizing such advancement. Thus, this is a pioneer study for struggling export-oriented manufacturing firms; managers of those firms can find the fundamental facts of progressing B2B tasks in order to gain profitability and overall economic stability. The academic literature in this aspect is likely to be enriched through this study. Overall, the research is the latest avenue of exploration in the export-oriented manufacturing firm's marketing analytics and AI aspects.

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Figures

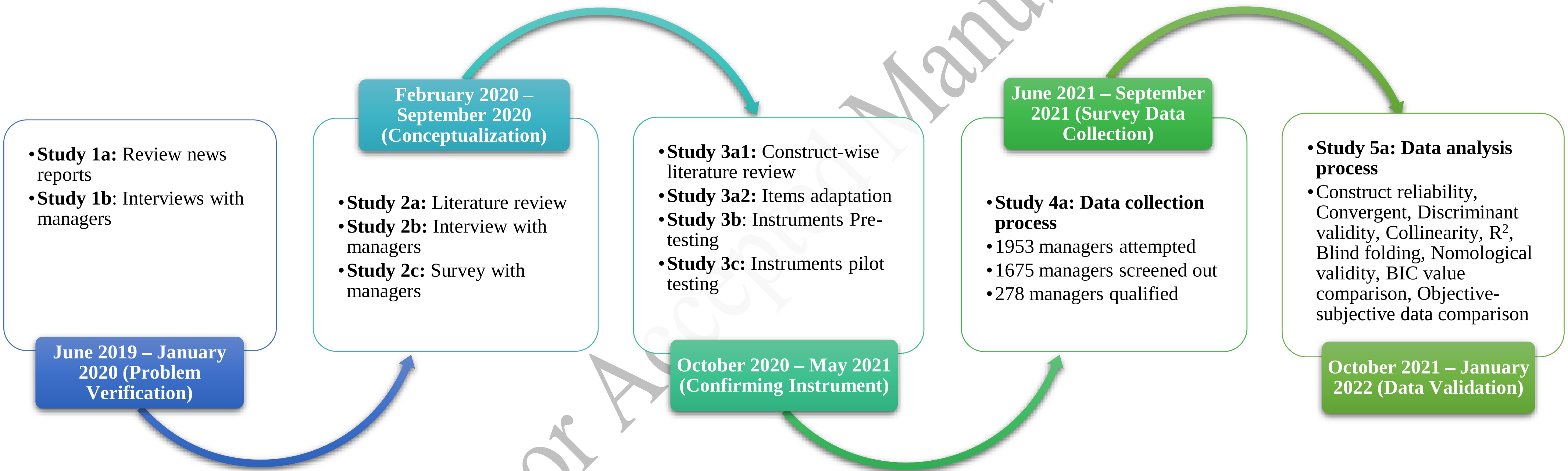


Figure 1. Research structure's flow chart

Figure 2. Word cloud of interviews

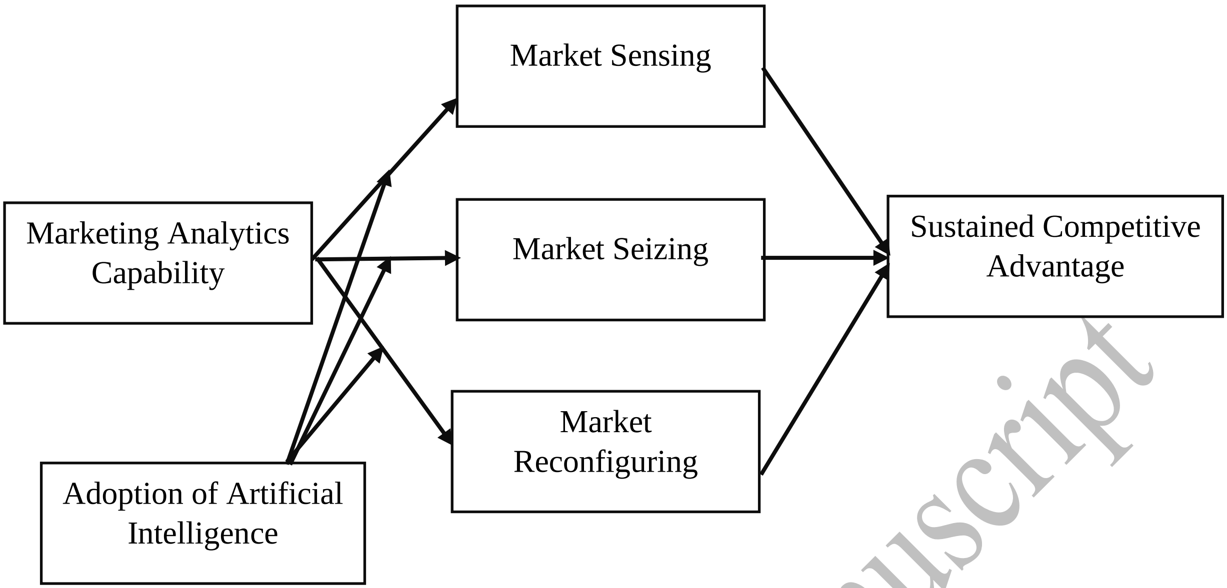
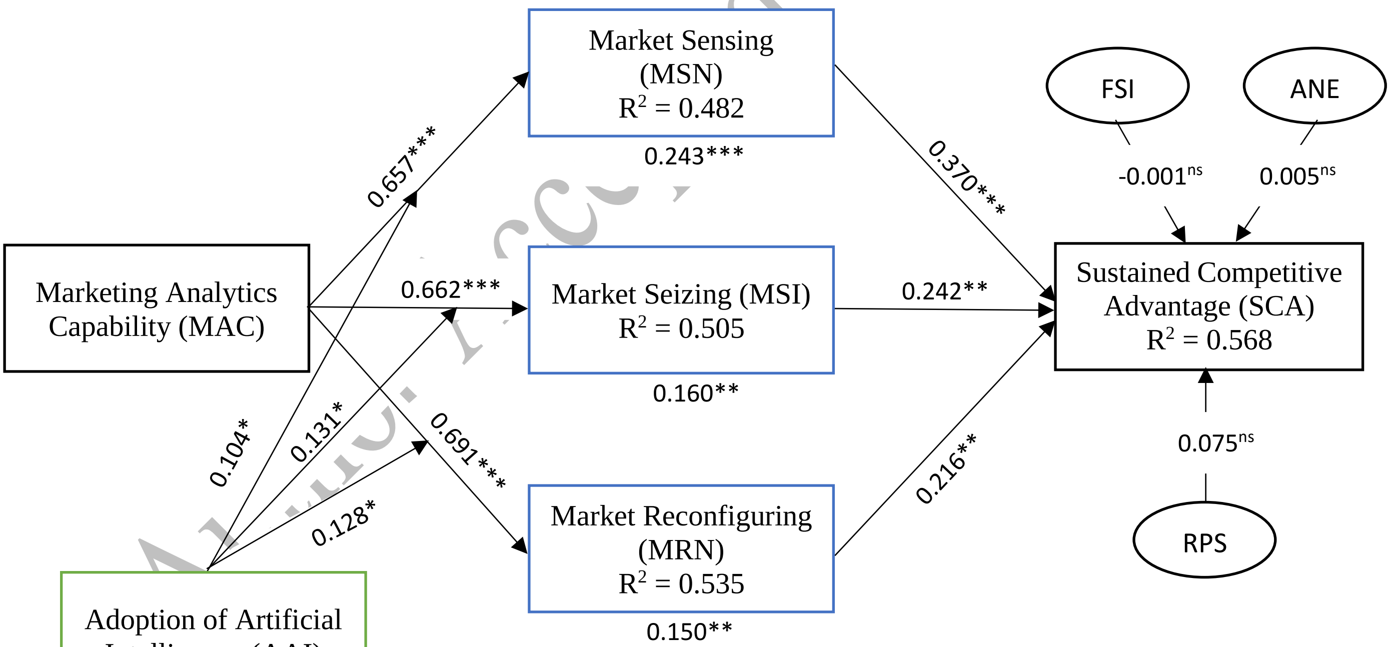


Figure 3. Conceptual model



Note: *p<0.05, **p<0.01, ***p<0.001, ns=not significant

Figure 4. Structural model

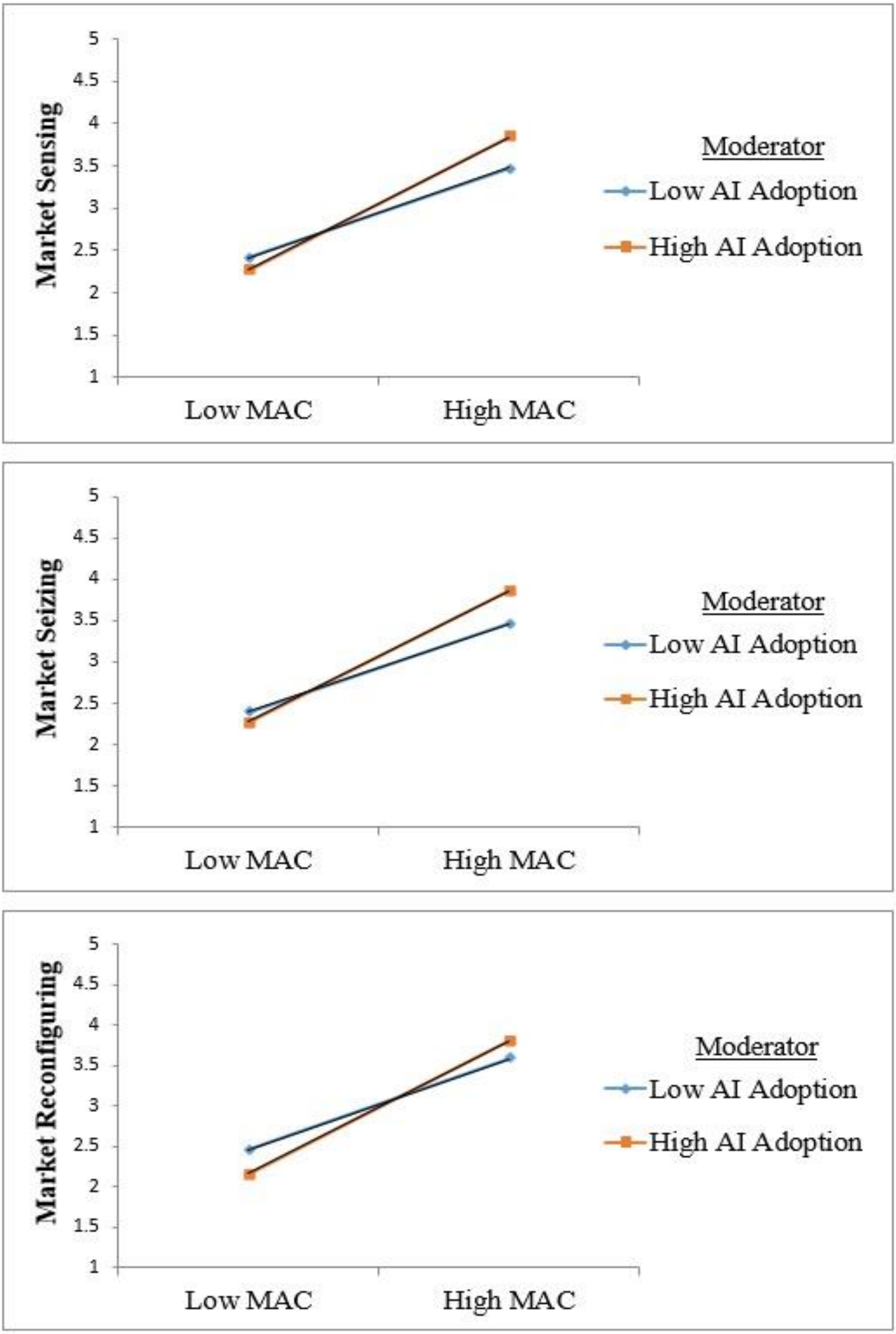


Figure 5. Moderation effect

Tables

Table 1. Demographic Profile

Items	Categories	%	Items	Categories	%
Managers Gender	Male	56.4	Managers Age	<25 Years	12.5
	Female	42.4		25 < 35 Years	37.7
	Do not disclose	1.2		35 < 45 Years	21.4
				45 ≤ Years	28.4
Size of Firm (Number of Employees)	≤ 50	25.7			
	51 ≤ 150	29.2	Managers Education	Diploma	18.3
	150 <	45.1		Undergrad	31.9
				Masters	41.2
Analytics Experience	< 3 Years	17.6		PhD	3.5
	3 ≤ 5 Years	41.2		Others	5.1
	5 < Years	41.2			

Table 2. Measurement items assessments.

Constructs	Items	Loadings	CR	AVE
Marketing Analytics Capability (MAC)	*Market-centred analytics facilitates our firm to:*		0.912	0.634
	MAC1: Understand buyers'/customers' changing demands.	0.747		
	MAC2: Personalize offerings based on buyers' desires.	0.811		
	MAC3: Create a distinct position meeting buyers' expectations.	0.824		
	MAC4: Generate various marketing mix models for allocating resources.	0.847		
	MAC5: Analyze holistic data that covers buyers', competitors', upstream/downstream suppliers', and overall market environment information.	0.780		
	MAC6: Protect buyers' and suppliers' privacy and security.	0.767		
Market Sensing (MSN)	*Using market-centric analytics capacity, we can:*		0.888	0.726
	MSN1: Detect competitors' strategies and tactics.	0.833		
	MSN2: Learn both micro and macro market environments.	0.873		
	MSN3: Identify and understand the market trend.	0.850		
Market Seizing (MSI)	*Using market-centric analytics capacity, we can:*		0.877	0.640
	MSI1: Find a buyer-centric solution.	0.786		
	MSI2: Adopt best practices to capture the market.	0.803		
	MSI3: Respond to any market imperfections.	0.807		
	MSI4: Seize every single opportunity from the market.	0.805		
Market Reconfiguring (MRN)	*Using market-centric analytics capacity, we can:*		0.869	0.689
	MRN1: Substantially renew relevant marketing processes.	0.782		
	MRN2: Constantly renew the ways of achieving our targets and objectives.	0.890		
	MRN3: Continuously implement new methods in marketing activities.	0.814		
Adoption of AI (AAI)	*Rate your response regarding AI adoption:*		0.890	0.670
	AAI1: AI adoption help firm to design product and services distinctly.	0.707		
	AAI2: AI can manage and optimize supply chains and reduce shipping costs and transit time.	0.913		
	AAI3: AI makes more accurate predictions of inventory demand and therefore reduces wastage.	0.866		
	AAI4: AI adoption accelerates the strategic decision process by measuring marketing predictions accurately.	0.773		
Sustained Competitive Advantage (SCA)	*Using market-centric analytics capacity, how well has your firm performed its _____________ goals relative to competitors in the last three years?*		0.925	0.755
	SCA1: Market share growth	0.892		
	SCA2: Sales growth	0.889		
	SCA3: Profitability	0.813		
	SCA4: Return on investment	0.880		

Table 3. Correlations and AVEs*

	AAI	MRN	MSI	MSN	MAC	SCA	ANE	RPS	FSI
AAI	0.819								
MRN	0.012	0.830							
MSI	0.097	0.372	0.800						
MSN	0.089	0.413	0.296	0.852					
MAC	0.079	0.318	0.396	0.385	0.797				
SCA	0.029	0.441	0.377	0.418	0.359	0.869			
ANE	0.015	0.089	0.088	0.089	0.110	0.076	N/A		
RPS	-0.065	-0.022	-0.030	0.042	0.020	0.080	-0.011	N/A	
FSI	0.025	0.267	0.160	0.185	0.147	0.169	0.353	0.061	N/A

**Square root of AVE on the diagonals.*

Table 4. Structural model's result

Hypotheses	Relationships	Path coefficients	Standard error	*t*-statistic
H1a	MAC → MSN	0.657	0.045	14.496
H1b	MAC → MSI	0.662	0.035	18.665
H1c	MAC → MRN	0.691	0.038	18.195
H2a	MSN → SCA	0.370	0.091	4.086
H2b	MSI → SCA	0.242	0.084	2.876
H2c	MRN → SCA	0.216	0.070	3.087
H3a	MAC → MSN → SCA	0.243	0.069	3.511
H3b	MAC → MSI → SCA	0.160	0.057	2.814
H3c	MAC → MRN → SCA	0.150	0.051	2.925
H4a	MAC*AAI moderating → MSN	0.104	0.053	1.983
H4b	MAC*AAI moderating → MSI	0.131	0.052	2.533
H4c	MAC*AAI moderating → MRN	0.128	0.050	2.576

Table 5. Future research stream and issues

Research stream	Future research questions
Marketing Analytics and AI in B2B context	- What metrics should a firm develop to understand buyers' changing demands? - What algorithms should a firm utilize to customize offerings based on buyers' desires? - How can a firm create a distinct position based on buyers' expectations in a data-rich environment? - In what capacity can a firm generate various marketing mix models for allocating resources? - What types of machine learning techniques should a firm utilize to detect the typical pattern from buyers', competitors', suppliers', and comprehensive market information? - How can a firm ensure the algorithmic techniques to protect buyers' and suppliers' data from misuse? - What metrics confirm the benchmark to detect competitors' strategies and tactics? - A firm can identify and understand the market trend in what optimal capacity of marketing analytics? - How can deep learning guide for responding to any market imperfections? - What capacity level can a firm utilize to renew relevant marketing processes substantially? - What are machine learning and deep learning's roles in distinctly designing B2B products and services offers? - How can AI mechanisms optimize supply chains reducing time and cost? - Which class of implications for marketing analytics and AI ensure the optimum level in strategic decision making?

Appendix

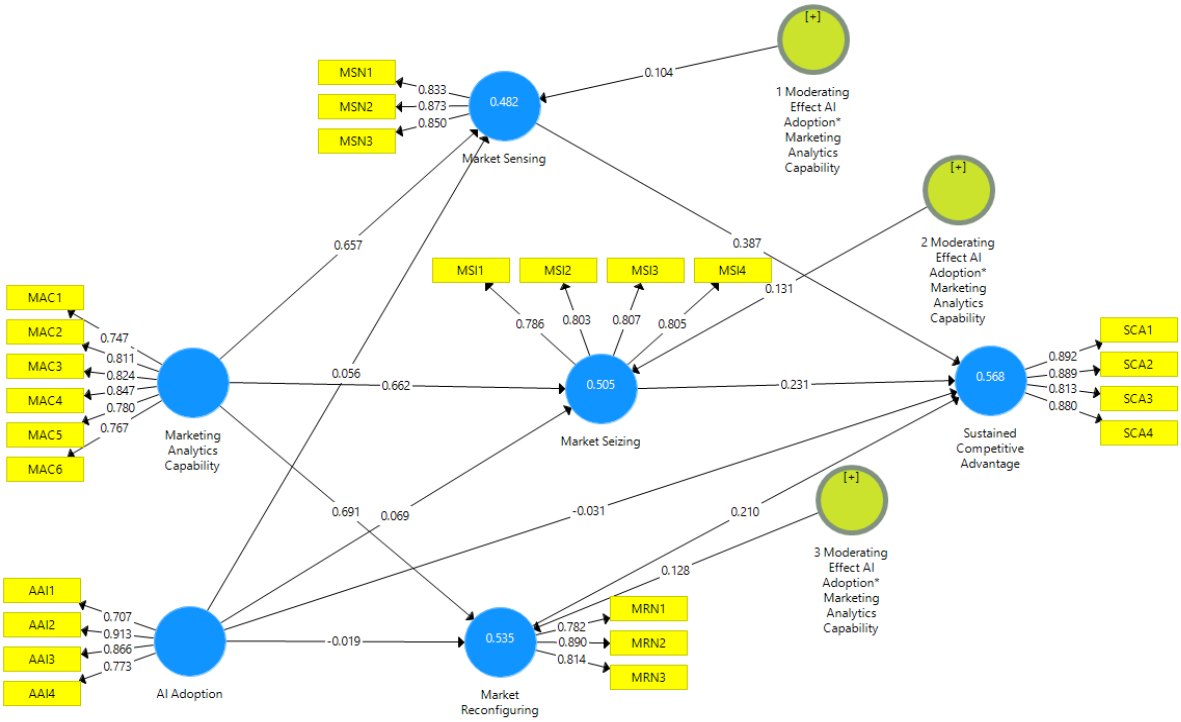


Figure 6. Measurement model from database

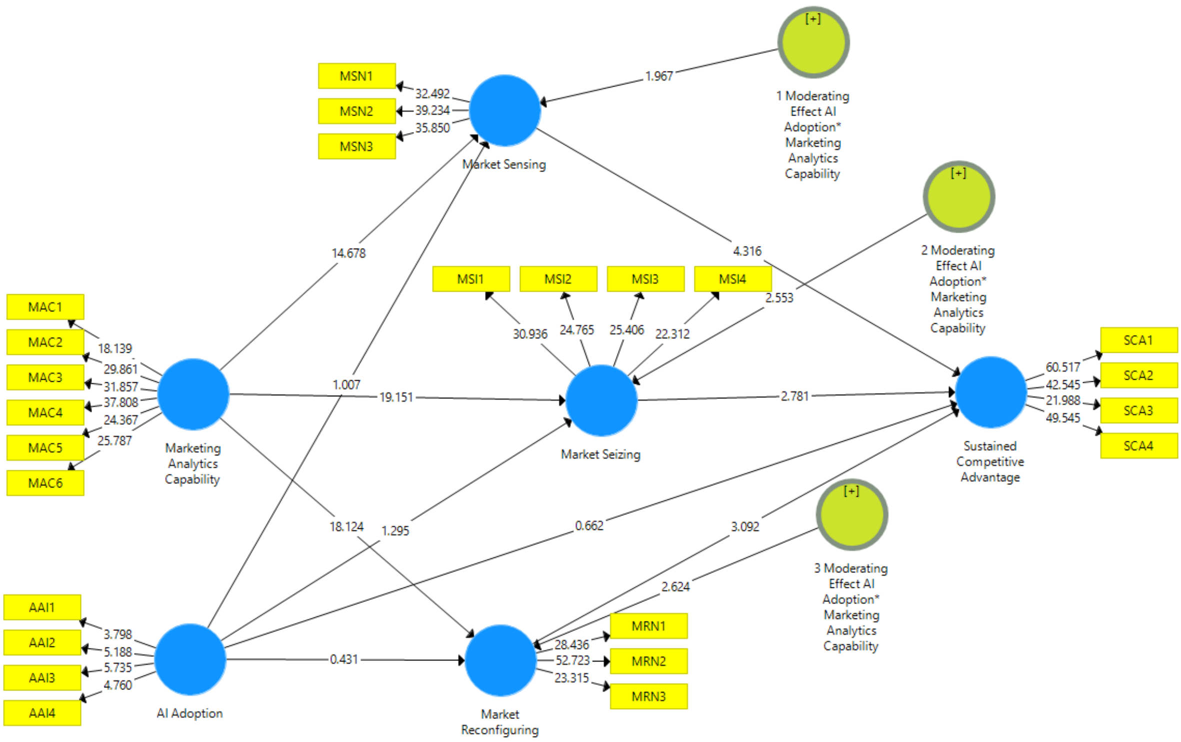


Figure 7. Structural model from database